

1. Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.

- **Initialize a repo:** `git init`
- **Stage all files:** `git add .`
- **Commit with a message:** `git commit -m "Your commit message"`
- **Push to remote:** `git push origin main` (*Note: replace 'main' with your branch name if different*)
- **Pull latest changes:** `git pull origin main`

2. What is the difference between git fetch and git pull? Answer in 2 lines.

- `git fetch` downloads the latest updates from the remote repository to your local system but does not merge them into your current working files.
- `git pull` combines `git fetch` and `git merge`, meaning it downloads the updates and immediately applies them to your current working branch.

3. Create a new GitHub repository, push any small text file from your local system, and share the repo link.

(You will need to do this on your own GitHub account. Here are the exact commands you can use in your terminal after clicking "New" on GitHub to create an empty repo):

Code:-

```
echo "This is a test file" > test.txt
```

```
git init
```

```
git add test.txt
```

```
git commit -m "Initial commit"
```

```
git branch -M main
```

```
git remote add origin https://github.com/YOUR-USERNAME/YOUR-REPO-NAME.git
```

```
git push -u origin main
```

Repo Link to share (replace with your actual details):

<https://github.com/YOUR-USERNAME/YOUR-REPO-NAME>

4. What is the difference between a Fork and a Pull Request? Answer in 2 lines.

- A **Fork** is a complete, personal copy of someone else's repository that lives on your own GitHub account so you can make changes safely.
- A **Pull Request (PR)** is a formal request asking the original repository's owner to review and merge your specific changes back into their main project.

5. Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used.

Code:-

1. Create and switch to the new branch

```
git checkout -b feature-test
```

2. (Make some changes to a file here, then stage and commit)

```
git add .
```

```
git commit -m "Made changes in feature-test branch"
```

3. Switch back to the main branch

```
git checkout main
```

4. Merge the feature branch into main

```
git merge feature-test
```